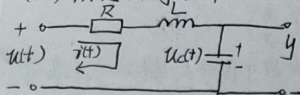


(一) 状态空间分析法



R-L-C 电路系统

$$u(t) = L \frac{di}{dt}$$

$$i = x_1, \quad u(t) = \frac{1}{C} \int i dt$$

$$\int i dt = x_2$$

$$\dot{x}_2 = x_1$$

$$x_1 = \dot{x}_1$$

列写微分方程

$$\begin{cases} iR + L \frac{di}{dt} + \frac{1}{C} \int i dt = u(t) \\ \frac{1}{C} \int i dt = y \end{cases}$$

$$\dot{x}_1 = \frac{di}{dt} = -\frac{R}{L} x_1 - \frac{1}{LC} x_2 - \frac{1}{L} u(t)$$

$$\dot{x}_2 = x_1$$

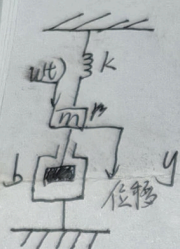
$$y = \frac{1}{C} \int x_1 dt = \frac{1}{C} \int \dot{x}_2 dt = \frac{1}{C} x_2$$

列写状态空间方程

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} -\frac{R}{L} & -\frac{1}{LC} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} -\frac{1}{L} \\ 0 \end{pmatrix} u$$

$$y = \begin{pmatrix} 0 & \frac{1}{C} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

机械系统的状态空间描述



$$kx \uparrow \uparrow kx$$

$$y = x_1$$

$$m\ddot{x} + b\dot{x} + kx = m\ddot{u}(t) - kx_1 - b\dot{x}_1$$

$$\dot{x}_2 = x_1$$

$$\dot{x}_1 = x_2$$

$$x_1 = x_2$$

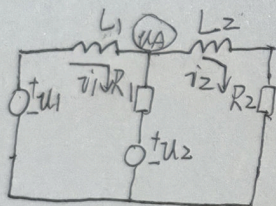
$$m\dot{x}_2 + kx_1 + b\dot{x}_2 = m\ddot{u}(t)$$

$$\dot{x}_2 = \frac{k}{m} x_1 + \dot{x}_2 + \frac{b}{m} \dot{x}_2 = \ddot{u}(t)$$

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{k}{m} x_1 - \frac{b}{m} x_2 + \ddot{u}(t) \\ y = x_1 \end{cases}$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \ddot{u}(t)$$

$$y = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



u_1, u_2 为输入量,
 u_A 为输出量

$$i_1 = x_1$$

$$i_2 = x_2$$

$$u_1 = L_1 \frac{di_1}{dt} + (i_1 - i_2) R_1 + u_2$$

$$u_2 = (i_1 - i_2) R_1 + L_2 \frac{di_2}{dt} + i_2 R_2$$

$$u_A = (i_1 - i_2) R_1 + u_2$$

$$u_1 = L_1 \dot{x}_1 + (x_1 - x_2) R_1 + u_2$$

$$u_2 = -(x_1 - x_2) R_1 + L_2 \dot{x}_2 + x_2 R_2$$

$$\dot{x}_1 = -\frac{R_1}{L_1} (x_1 - x_2) + \frac{1}{L_1} (u_1 - u_2)$$

$$\dot{x}_2 = \frac{R_1}{L_2} (x_1 - x_2) - \frac{R_2}{L_2} x_2 + \frac{1}{L_2} u_2$$

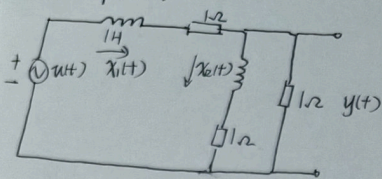
$$= \frac{R_1}{L_2} x_1 - \left(\frac{R_1}{L_2} + \frac{R_2}{L_2} \right) x_2 + \frac{1}{L_2} u_2$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} -\frac{R_1}{L_1} & \frac{R_1}{L_1} \\ \frac{R_1}{L_2} & -\frac{R_1 + R_2}{L_2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} \frac{1}{L_1} & -\frac{1}{L_1} \\ 0 & \frac{1}{L_2} \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

$$y = \begin{pmatrix} 0 & -R_1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$$

(= 例 2.3. 求取该电路在 $t=0$ 时刻的初始值为

求单位阶跃



$$\begin{aligned} u(t) &= \dot{x}_1 + x_1 + \cancel{x_2} - \cancel{x_2} (x_1 - x_2) \Rightarrow \dot{x}_1 = -2x_1 + x_2 + u(t) \\ \dot{x}_2 + x_2 &= x_1 - x_2 \Rightarrow \dot{x}_2 = x_1 - 2x_2 \\ y(t) &= x_1 - x_2 = \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u$$

(= 例 2.3. 线性定常系统的齐次状态方程为

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u \quad x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

求单位阶跃输入作用下的状态响应。 $h[u(t)] = \frac{1}{s}$

$$\dot{X} = AX + bu$$

左右同时拉氏变换

$$sX(s) - X(0) = AX(s) + bU(s)$$

$$sX(s) = AX(s) + X(0) + bU(s)$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \cdot \frac{1}{s} = \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix}$$

$$(sI - A) \cdot X(s) = \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix}$$

$$X(s) = \left(\frac{1}{s} \right) (sI - A)^{-1} \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix} \left[\begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} \right]^{-1} \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix} \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}^{-1} \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix}$$

$$= \frac{1}{(s+1)(s+2)} \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} \begin{pmatrix} 1 \\ \frac{1}{s} \end{pmatrix}$$

求系统的响应

所求 $\rightarrow \frac{1}{s}$
阶跃 $\rightarrow 1$

$$X(t) = \mathcal{L}^{-1}[(sI - A)^{-1}]X(0) + \mathcal{L}^{-1}[(sI - A)^{-1}bU(s)]$$

$$\frac{1}{s(s+1)+2} = \frac{1}{s^2+3s+2} = \frac{1}{(s+1)(s+2)}$$

$$= \left[\frac{\frac{s+2}{(s+1)(s+2)}}{(s+1)(s+2)} + \frac{\frac{1}{(s+1)(s+2)}}{(s+1)(s+2)} \right]$$

$$= \left[\frac{-1}{(s+1)(s+2)} \right]$$

$$\frac{A}{s+1} + \frac{B}{s+2} : A(s+2) + B(s+1) = s+3 \quad \begin{cases} A+B=1 \\ 2A+B=3 \end{cases} \quad \begin{cases} A=2 \\ B=-1 \end{cases}$$

$$\mathcal{L}^{-1} \left[\frac{-1}{(s+1)(s+2)} \right] = \mathcal{L}^{-1} \left[\frac{2}{s+1} - \frac{1}{s+2} \right] = 2e^{-t} - e^{-2t}$$

$$\text{Res}[F(s)e^{st}, 0] = \lim_{s \rightarrow 0} s \frac{e^{st}}{s(s+1)(s+2)} = \frac{1}{2}$$

$$\text{Res}[F(s)e^{st}, -1] = \lim_{s \rightarrow -1} (s+1) \frac{e^{st}}{s(s+1)(s+2)} = e^{-t}$$

$$\text{Res}[F(s)e^{st}, -2] = \lim_{s \rightarrow -2} (s+2) \frac{e^{st}}{s(s+1)(s+2)} = \frac{1}{2}e^{-2t}$$

$$\mathcal{L}^{-1} \left[\frac{1}{(s+1)(s+2)} \right]$$

$$= \text{Res}[F(s)e^{st}, -1] = \lim_{s \rightarrow -1} \frac{-e^{st}}{s+2} = -e^{-t}$$

$$+ \text{Res}[F(s)e^{st}, -2] = \lim_{s \rightarrow -2} \frac{-e^{st}}{s+1} = -\frac{1}{2}e^{-2t}$$

$$= -e^{-t} + \frac{1}{2}e^{-2t}$$

$$\mathcal{L}^{-1}[X(s)] = \begin{pmatrix} 2e^{-t} - e^{-2t} + \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-t} \\ -e^{-t} + e^{-2t} \end{pmatrix}$$

$$= \begin{pmatrix} e^{-t} - \frac{1}{2}e^{-2t} + \frac{1}{2} \\ -e^{-t} + e^{-2t} \end{pmatrix}$$

(二) 求状态转移矩阵, 使用 $\Phi(t) = e^{-tA} = \mathcal{L}^{-1}[(sI - A)^{-1}]$

例 2.9 $\dot{x} = Ax + bu$

$$A = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix} \quad b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\Phi(t) = e^{-tA} = \mathcal{L}^{-1}[(sI - A)^{-1}]$$

$$sI - A = \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} - \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} s+2 & -1 \\ -1 & s+2 \end{pmatrix}$$

$$sI - A = \begin{pmatrix} s+2 & -1 \\ -1 & s+2 \end{pmatrix} - \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} s+4 & -2 \\ -2 & s+4 \end{pmatrix}$$

$$|sI - A| = (s+4)^2 - 4 = s^2 + 8s + 12 = (s+2)(s+6)$$

$$(sI - A)^{-1} = \frac{1}{|sI - A|} \begin{pmatrix} s+4 & 2 \\ 2 & s+4 \end{pmatrix}$$

$$= \frac{1}{(s+2)(s+6)} \begin{pmatrix} s+4 & 2 \\ 2 & s+4 \end{pmatrix} = \begin{pmatrix} \frac{s+4}{(s+2)(s+6)} & \frac{2}{(s+2)(s+6)} \\ \frac{2}{(s+2)(s+6)} & \frac{s+4}{(s+2)(s+6)} \end{pmatrix}$$

$$\mathcal{L}^{-1}[(sI - A)^{-1}]$$

$$\mathcal{L}^{-1}\left[\frac{s+4}{(s+2)(s+6)}\right] = \text{Res}[F(s)e^{st}, -2] = \lim_{s \rightarrow -2} \frac{e^{st}(s+4)}{(s+6)} = \frac{e^{-2t}}{2}$$

$$+ \text{Res}[F(s)e^{st}, -6] = \lim_{s \rightarrow -6} \frac{e^{st}(s+4)}{(s+2)} = \frac{e^{-6t}}{2}$$

$$\mathcal{L}^{-1}\left[\frac{2}{(s+2)(s+6)}\right] = \text{Res}[F(s)e^{st}, -2] + \text{Res}[F(s)e^{st}, -6]$$

$$\text{Res}[F(s)e^{st}, -2] = \lim_{s \rightarrow -2} \frac{2e^{st}}{s+6} = \frac{1}{2}e^{-2t}$$

$$\text{Res}[F(s)e^{st}, -6] = \lim_{s \rightarrow -6} \frac{2e^{st}}{s+2} = -\frac{1}{2}e^{-6t}$$

$$\text{得 } \mathcal{L}^{-1}\left[\frac{2}{(s+2)(s+6)}\right] = \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-6t}$$

$$\text{故 } \Phi(t) = \begin{pmatrix} \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-6t} & \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-6t} \\ \frac{1}{2}e^{-2t} - \frac{1}{2}e^{-6t} & \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-6t} \end{pmatrix}$$

例 2.12 $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$ 用线性变换求状态转移矩阵

$$\Phi(t) = e^{-tA} = \mathcal{L}^{-1}[(sI - A)^{-1}]$$

$$sI - A = \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} = \begin{pmatrix} s & -1 \\ 2 & s+3 \end{pmatrix}$$

$$(sI - A)^{-1} = \frac{1}{|sI - A|} \cdot (sI - A)^*$$

$$|sI - A| = s^2 + 3s + 2 = (s+2)(s+1)$$

$$(sI - A)^* = \begin{pmatrix} s+3 & 1 \\ 2 & s \end{pmatrix} \quad (sI - A)^{-1} = \frac{1}{(s+1)(s+2)} \begin{pmatrix} s+3 & 1 \\ 2 & s \end{pmatrix}$$

$$\text{则 } (sI - A)^{-1} = \begin{pmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{pmatrix}$$

$$\begin{aligned} \mathcal{L}^{-1}\left[\frac{s+3}{(s+1)(s+2)}\right] &= \text{Res}[F(s)e^{st}, -1] + \text{Res}[F(s)e^{st}, -2] \\ &= \lim_{s \rightarrow -1} \frac{(s+3)e^{st}}{s+2} + \lim_{s \rightarrow -2} \frac{(s+3)e^{st}}{s+1} \\ &= 2e^{-t} + -e^{-2t} = 2e^{-t} - e^{-2t} \end{aligned}$$

$$\mathcal{L}^{-1}\left[\frac{1}{(s+1)(s+2)}\right] = e^{-t} + (e^{-2t}) = e^{-t} - e^{-2t}$$

$$\mathcal{L}^{-1}\left[\frac{2}{(s+1)(s+2)}\right] = 2e^{-t} + 2e^{-2t}$$

$$\mathcal{L}^{-1}\left[\frac{s}{(s+1)(s+2)}\right] = -e^{-t} + 2e^{-2t}$$

$$\text{则 } \Phi(t) = \begin{pmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ 2e^{-t} - e^{-2t} & -e^{-t} + 2e^{-2t} \end{pmatrix}$$

$$\dot{x} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 1 & -4 & 3 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u, y = (1 \ -1 \ 1) x$$

进行能控性结构分解。

① 判断系统的能控性。构造矩阵 $M = (b \ Ab \ A^2b)$

$$b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad Ab = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 1 & -4 & 3 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} \quad A^2b = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 0 \\ 1 & -4 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 & 8 & -4 \\ 0 & 1 & 0 \\ 4 & -14 & 8 \end{pmatrix} \quad A^2b = \begin{pmatrix} -4 \\ 0 \\ 8 \end{pmatrix}$$

$$M = \begin{pmatrix} 0 & -1 & 4 \\ 0 & 0 & 0 \\ 1 & 3 & 8 \end{pmatrix} = 2 < 3 = n. \text{ 则不能控}$$

② 选取能控子矩阵前两列，再补充线性无关列 $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ 为 $P_c = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 3 & 0 \end{pmatrix}$. $P_c^{-1} = -\begin{pmatrix} 3 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

③ 对原系统进行线性变换 $\tilde{A}_c = P_c^{-1} A P_c$, $\tilde{b}_c = P_c^{-1} b$, $\tilde{c}_c = c P_c$ 结构分解: 能控 能观
 $\tilde{A}_c = \begin{pmatrix} 0 & -4 & 2 \\ 1 & -4 & -2 \\ 0 & 0 & 1 \end{pmatrix}$ $\tilde{b}_c = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$, $\tilde{c}_c = (1 \ 2 \ -1)$ P_c 取前两列 $M = (b \ Ab \ A^2b)$ 前两列 $N = \begin{pmatrix} c \\ cA \\ cA^2 \end{pmatrix}$
 $\tilde{A} = \begin{pmatrix} \tilde{A}_{11} & \tilde{A}_{12} \\ 0 & \tilde{A}_{22} \end{pmatrix}$ $\tilde{b} = \begin{pmatrix} \tilde{b}_1 \\ 0 \end{pmatrix}$ $\tilde{c} = (\tilde{c}_1 \ \tilde{c}_2)$
 $\tilde{A} = \begin{pmatrix} \tilde{A}_{11} & 0 \\ \tilde{A}_{21} & \tilde{A}_{22} \end{pmatrix}$ $\tilde{b} = \begin{pmatrix} \tilde{b}_1 \\ \tilde{b}_2 \end{pmatrix}$ $\tilde{c} = (\tilde{c}_1 \ 0)$
 进行能控性结构分解 $\begin{pmatrix} \tilde{x}_c \\ \tilde{x}_c \end{pmatrix} = \tilde{A}_c \begin{pmatrix} \tilde{x}_c \\ \tilde{x}_c \end{pmatrix} + \tilde{b}_c u, y = \tilde{c}_c x$

$$\dot{\tilde{x}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -4 & 3 \end{pmatrix} \tilde{x} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u,$$

$y = (1 \ 1 \ 0) \tilde{x}$. 能观测性判别, 若不能, 按能观测性进行结构分解。

例4-12 控制系统框图如图4-2所示,要求系统渐近稳定,试确定增益 \$k\$ 的取值范围。

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & -2 & 1 \\ -k & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ k \end{pmatrix} r$$

输入为0, 状态空间的原点为系统的平衡状态。

取 \$Q\$ 为 正定对称矩阵

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

为了简化计算: \$V(x)\$ 沿任意轨迹并不等于零的原因是,

$$\dot{V}(x) = -x^T Q x = -x_3^2$$

\$\dot{V}(x) = 0\$ 的条件 \$x_3 = 0\$, 从状态方程可看出若 \$x_3 = 0\$, \$x_3 = 0\$, 则 \$x_1\$ 和 \$x_2\$ 也必须为零。因此 \$\dot{V}(x)\$ 恒为零的情况只在原点成立, 故 \$Q\$ 可取单位阵, 简化计算。

4-2 试确定下列二次型函数的符号(正定性)

$$V(x) = x_1^2 + 4x_1x_2 + 5x_2^2 - 2x_1x_3 + x_3^2$$

$$V(x) = x^T P x$$

$$= \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= P_{11}x_1^2 + 2P_{12}x_1x_2 + P_{22}x_2^2$$

$$V(x) = x^T P x$$

$$= \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} P_{11} & P_{12} & P_{13} \\ P_{12} & P_{22} & P_{23} \\ P_{13} & P_{23} & P_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$P_{11} = 1, 2P_{12} = 4, P_{22} = 5, 2P_{23} = -2, P_{33} = 1$$

$$P = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 5 & -1 \\ 0 & -1 & 1 \end{pmatrix}$$

$$\Delta_1 = P_{11} = 1 > 0$$

$$\Delta_2 = \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = 1 > 0$$

$$\Delta_3 = \begin{vmatrix} 1 & 2 & 0 \\ 2 & 5 & -1 \\ 0 & -1 & 1 \end{vmatrix} = (5-1) - 2 \times 2 = 0$$

为半正定

例3-4. 把下面系统的状态空间方程转换为能控标准型

4-5 用李雅普诺夫第二法判断线性系统 $\dot{x}_1 = -x_1 + x_2, \dot{x}_2 = 2x_1 - 3x_2$ 平衡状态的稳定性

解法. 取 $Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $P = \begin{pmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{pmatrix}$, $A = \begin{pmatrix} -1 & 1 \\ 2 & -3 \end{pmatrix}$

$$-Q = A^T P + P A$$

$$= \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{pmatrix} + \begin{pmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 2 & -3 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -2P_{11} + 4P_{12} & -P_{12} + 2P_{22} \\ -3P_{12} - P_{22} & 3P_{12} - 3P_{22} \end{pmatrix} = \begin{pmatrix} -2P_{11} + 4P_{12} & -4P_{12} + 2P_{22} \\ -4P_{12} + 2P_{22} & -6P_{22} \end{pmatrix}$$

$$P_{11} = \frac{2}{3}, P_{12} = \frac{1}{12}, P_{22} = \frac{1}{6}$$

$$P = \begin{pmatrix} \frac{2}{3} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{6} \end{pmatrix}, \Delta_1 = \frac{2}{3} > 0$$

$$\Delta_2 = \begin{vmatrix} \frac{2}{3} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{6} \end{vmatrix} = \frac{2}{3} \times \frac{1}{6} - \frac{1}{144} > 0$$

则在平衡状态(原点)大范围渐近稳定的.

$$V(x) = x^T P x$$

$$= (x_1 \ x_2) \begin{pmatrix} \frac{2}{3} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{6} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= \left(\frac{2}{3}x_1 + \frac{1}{12}x_2 \quad \frac{1}{12}x_1 + \frac{1}{6}x_2 \right) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= \frac{2}{3}x_1^2 + \frac{1}{6}x_1x_2 + \frac{1}{6}x_1x_2 + \frac{1}{6}x_2^2 = \frac{2}{3}x_1^2 + \frac{1}{3}x_1x_2 + \frac{1}{6}x_2^2$$

4-7 X. 此题为判据 -

例3-4. 把下面系统的状态空间方程, 转换为能控标准型

$$\begin{cases} \dot{x} = Ax + bu \\ y = cx \end{cases}$$

$$A = \begin{pmatrix} 0 & 2 & -2 \\ 1 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}, b = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^T$$

判断系统的能控性

$$M = (b, Ab, A^2b) = \begin{pmatrix} 2 & 0 & -4 \\ 1 & 1 & -5 \\ 1 & 3 & 1 \end{pmatrix} \quad \text{rank } M = 3, \text{ 能控}$$

$$W(s) = c(sI - A)^{-1}b$$

$$sI - A = \begin{pmatrix} s & -2 & 2 \\ -1 & s-1 & 2 \\ -2 & 2 & s-1 \end{pmatrix}$$

$$|sI - A| = s \begin{vmatrix} s-1 & 2 \\ 2 & s-1 \end{vmatrix} + (-1)^{2+1} \begin{vmatrix} -2 & 2 \\ 2 & s-1 \end{vmatrix} + (-1)^{3+1} \begin{vmatrix} -2 & 2 \\ -1 & s-1 \end{vmatrix}$$

$$= s[(s-1)^2 - 4] + (-2s+2-4) - 2(-4-2s+2)$$

$$= s[s^2 - 2s - 3] - 2s - 2 + 4s + 4$$

$$= s^3 - 2s^2 - s + 2$$

$$(sI - A)^* = \begin{pmatrix} s^2 - 2s - 3 & 2s + 2 & -2s - 2 \\ s - 5 & s^2 - s + 4 & 2s - 2 \\ 2s - 4 & -2s + 4 & s^2 - s - 2 \end{pmatrix}$$

$$a_{11} = (s-1)^2 - 4 = s^2 - 2s - 3$$

$$a_{12} = -1 \times [1 - s + 4] = s - 5$$

$$a_{13} = -2 + 2s - 2 = 2s - 4$$

$$a_{21} = -1 \times [-2s + 2 - 4] = 2s + 2$$

$$a_{22} = s^2 - s + 4$$

$$a_{23} = -(-s - 4) = s + 4$$

$$a_{31} = -4 - 2(s-1) = -2s - 2$$

$$a_{32} = -(2s + 2) = -2s - 2$$

$$a_{33} = s(s-1) - 2 = s^2 - s - 2$$

$$(sI - A)^* = \begin{pmatrix} s^2 - 2s - 3 & 2s + 2 & -2s - 2 \\ s - 5 & s^2 - s + 4 & 2s - 2 \\ 2s - 4 & -2s + 4 & s^2 - s - 2 \end{pmatrix}$$

$$(sI - A)^{-1} = \frac{1}{|sI - A|} (sI - A)^*$$

$$W(s) = \frac{1}{s^3 - 2s^2 - s + 2} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} s^2 - 2s - 3 & 2s + 2 & -2s - 2 \\ s - 5 & s^2 - s + 4 & 2s - 2 \\ 2s - 4 & -2s + 4 & s^2 - s - 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$$

$$= \frac{4s^2 - 4s - 16}{s^3 - 2s^2 - s + 2}$$

$$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} s^2 - 2s - 3 & 2s + 2 & -2s - 2 \\ s - 5 & s^2 - s + 4 & 2s - 2 \\ 2s - 4 & -2s + 4 & s^2 - s - 2 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2s^2 + 3s^2 + 2s - s - 5s \\ -24 + 10 - 2 \end{pmatrix} = 4s^2 - 4s - 16$$

$$\text{① 传递函数法 } W(s) = c(sI - A)^{-1}b$$

$$\text{② } T_c = (0 \ 0 \ 1) M^{-1}, \quad T_c^{-1} = \begin{pmatrix} t_{c1} \\ t_{c2} \\ t_{c3} \end{pmatrix}$$

$$t_{c1} = M^{-1} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad T_c^{-1} = (t_{c1}, t_{c2}, t_{c3})$$

$$W(s) = c(sI - A)^{-1}b = V$$

$$b = (sI - A)^{-1}V$$

$$\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = (sI - A)^{-1}V$$

$$V = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$v_1 = \begin{vmatrix} 2 & -2 & 2 \\ -1 & s-1 & 2 \\ 1 & 2 & s-1 \end{vmatrix} / \det(sI - A)$$

$$v_2 = \begin{vmatrix} s & 2 & 2 \\ -1 & 1 & 2 \\ -2 & 1 & s-1 \end{vmatrix} / \det(sI - A)$$

$$v_3 = \begin{vmatrix} s & -2 & 2 \\ -1 & s-1 & 2 \\ -2 & 2 & s-1 \end{vmatrix} / \det(sI - A)$$

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$C = \begin{pmatrix} -a_0 & -a_1 & -a_2 \\ b_0 & b_1 & b_2 \end{pmatrix} = \begin{pmatrix} -16 & -4 & 4 \end{pmatrix}$$